

SCOTT–VOGELIUS–NITSCHKE ON A POLYGONALLY APPROXIMATED BOUNDARY: CONVERGENCE, THE MISSING PRESSURE TRACTION, AND THE PENALTY THRESHOLD

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ABSTRACT. We analyse the Scott–Vogelius–Nitsche method of Gjerde and Scott for two-dimensional Stokes flow whose no-slip curve is replaced by an inscribed polygon: continuous P_k velocities ($k \geq 4$) that are exactly divergence-free, with Dirichlet data imposed weakly on the polygon by Nitsche’s method. This answers Scott’s zero-gradient prize question, which asks whether the H^1 error behaves like $h_\Gamma^{3/2} + h_\Omega^k$ for general Stokes data and, if not, why not.

For the method as printed, the answer is no as soon as the pressure is not constant on the boundary. The printed Nitsche form omits the traction $-pn$. We prove that the method converges, at the sharp rate h/μ in the H^1 seminorm and at the exact rate $(h/\mu)^{1/2}$ in the mesh-dependent energy norm. To leading order the error is an explicit field: the discrete velocity leaks through the wall with normal velocity $(h/\mu)(p - \bar{p})$, with constant exactly 1 in the slip and energy norms. The H^1 rate is 1 and not 1/2 only because incompressibility turns the normal part of the missing traction into a tangential derivative.

The naive pressure-consistent correction is singular. A mean-corrected variant is well posed and attains $h_\Gamma^{3/2} + h_\Omega^k$ for general data and every fixed $k \geq 4$, provided the boundary edges are not too short relative to the bulk mesh, using the Guzmán–Scott inf-sup constant made uniform over the perturbed domains.

We also show that the Nitsche penalty must scale like $\rho = h_\Omega / \min_e |e|$, with necessity certified in exact rational arithmetic. A computational study up to $N = 256$ boundary chords confirms every rate and constant. It includes a pure-pressure test that isolates the traction response exactly, and a penalty study in which the threshold is linear in ρ with slope about 20.

1. INTRODUCTION

Exactly divergence-free finite elements are sensitive to how a curved no-slip boundary is represented. On a polygon, a conforming, piecewise-smooth, divergence-free field that vanishes on the boundary satisfies extra homogeneous constraints on its gradient at the boundary vertices. When a curved wall is replaced by an inscribed polygon with more and more vertices, these constraints cause order-one gradient errors at the vertices. The H^1 error of strongly imposed Scott–Vogelius elements then decays only like $h^{1/2}$ [9, 4].

Gjerde and Scott [4] impose the Dirichlet condition on the polygon weakly, by Nitsche’s method [8], and observed H^1 errors of order $h^{3/2}$ on a benchmark with constant pressure. Scott’s zero-gradient prize [9] asks for a proof that Scott–Vogelius–Nitsche converges for general Stokes data, including a computational study of the Nitsche penalty. It states the expected error $h_\Gamma^{3/2} + h_\Omega^k$ and adds: *if this does not hold, why not?*

Answer. For the method as printed in [4, (27)–(28)], the expected estimate holds if and only if the pressure is constant on the wall. The printed Nitsche form contains no traction term $-pn$. When p varies along Γ , the penalty, not a traction, balances the pressure. The discrete velocity

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then leaks through the wall with normal velocity $(h/\mu)(p - \bar{p})$, and the error is dominated by this leak (Theorems A–C and Corollary B'). A pressure-consistent correction restores the expected rate (Theorem D), and the penalty threshold must grow with the ratio of bulk to boundary mesh size (Proposition E).

Main results. The notation is fixed in Section 2. Throughout, $h = h_\Omega$ is the bulk mesh size used in Scott's penalty μ/h , h_Γ is the boundary edge length, $\gamma = \mu h_\Gamma/h$ and $G = \|p - \bar{p}\|_{L^2(\Gamma)}$. Constants do not depend on h, h_Γ, μ, γ or ρ .

Theorem A (Theorem 6.1). *The printed method GS converges, and its error in the mesh-dependent energy norm is*

$$\|\tilde{u} - u_h\| \asymp (h/\mu)^{1/2} G + O((1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h),$$

with matching upper and lower bounds.

Theorem C (Theorem 6.4). $\|\nabla(\tilde{u} - u_h)\| \leq C_p h/\mu + C((1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h)$. *The rate 1 (rather than $1/2$) holds because $\operatorname{div} v = 0$ turns the normal load into a tangential derivative (Lemma 6.3).*

Theorem B (Theorem 6.6). *In the regime where the pressure term dominates, the slip satisfies $\|\tilde{u} - u_h\|_{L^2(\Gamma_h)} \asymp (h/\mu)G$ and $\|\nabla(\tilde{u} - u_h)\| \asymp h/\mu$. So Theorem C is sharp.*

Corollary B' (Corollary 6.7). $\tilde{u} - u_h = (h/\mu)v_\varphi + o((h/\mu)^{1/2})$ in the energy norm, where v_φ is an explicit divergence-free field with $v_\varphi \approx -(p - \bar{p})n$ on Γ_h . In particular

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1.$$

Theorem D (Theorem 7.2). *The naive consistent form, which adds $\langle p_h, v \cdot n \rangle$, is singular (Proposition 7.1). The corrected method CNS*, which adds $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$, is well posed for $\gamma \geq \gamma_2$ and satisfies*

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$$

for general Stokes data, every fixed $k \geq 4$ and bounded γ , provided the boundary edges are not too short relative to the bulk ($h_\Gamma \geq h_\Omega^4$ for even k , $h_\Gamma \geq h_\Omega^2$ for odd k). This is Scott's expected rate. Without that proviso a single lower-order term, due to the net flux of the interpolated outer data, remains.

Proposition E (Proposition 8.1). *The coercivity threshold of the Nitsche penalty scales like $\rho = h/\min_{e \in \mathcal{E}_h^\Gamma} |e|$. The condition $\mu \geq 4\kappa C_I^2 \rho$ is sufficient. The condition $\mu \gtrsim \rho$ is necessary on meshes whose boundary stars are close to two reference stars, certified in exact rational arithmetic.*

Theorem A with $G = 0$ gives $\|\tilde{u} - u_h\| \leq C((1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h)$. This recovers, for the printed method, the constant-pressure benchmark studied numerically in [4] and analysed in [2], which covers $k \geq 4$ and also Clough–Tocher splits with $k \geq 2$. Our proof does not use [2]. All results hold for every fixed $k \geq 4$; the computations use $k = 4$.

Computations. Section 9 reports $k = 4$ computations with up to $N = 256$ boundary chords. They show the following.

- The CNS* H^1 rate decreases monotonically to $3/2$, independently of the pressure.
- The printed method has rate 1 whenever $G > 0$.
- On a pure-pressure test that isolates the traction response exactly, the slip and energy constants of Corollary B' converge to 1.
- In the penalty study, the coercivity threshold μ^* is linear in ρ over more than a decade, with $\mu^*/\rho \approx 18$ –21.

All code, raw outputs and logs are in the accompanying repository (Appendix A).

Organisation. Section 2 fixes the setting. Sections 3–5 collect geometric and discrete tools and the error equation. Section 6 treats the printed method, Section 7 the pressure-consistent method, Section 8 the penalty threshold and Section 9 the computations.

2. SETTING, METHODS AND ASSUMPTIONS

Domain. Let $R = (-L, L)^2$ with $L > 1$, let D be the open unit disk, $\Omega = R \setminus \overline{D}$ and $\Gamma = \partial D$. Let P_h be a convex polygon with vertices on Γ , $\Gamma_h = \partial P_h$ and $\Omega_h = R \setminus P_h$. Since $P_h \subset D$, we have $\Omega \subset \Omega_h$. The strip $S_h = \Omega_h \setminus \overline{\Omega}$ lies between each chord and its arc.

Mesh. $\mathcal{T}_h$ is a conforming triangulation of Ω_h , and $\mathcal{E}_h^\Gamma$ is its set of edges on Γ_h . We use the parameters

$$h = h_\Omega := \max_{T \in \mathcal{T}_h} \text{diam } T, \quad h_\Gamma := \max_{e \in \mathcal{E}_h^\Gamma} |e|, \quad \rho := \frac{h}{\min_{e \in \mathcal{E}_h^\Gamma} |e|}, \quad \gamma := \frac{\mu h_\Gamma}{h}.$$

Here h is the h in Scott's penalty μ/h , and γ is the effective boundary penalty, so that $\mu/h = \gamma/h_\Gamma$.

Assumption 2.1.

- (M0) Shape regularity, and $c_0 h_\Gamma \leq |e| \leq h_\Gamma$ for all $e \in \mathcal{E}_h^\Gamma$. Hence $\rho/h \leq 1/(c_0 h_\Gamma)$. No quasi-uniformity is assumed; $h_\Gamma \ll h$ is allowed.
- (M1) No vertex of $\mathcal{T}_h$ is singular. A vertex is singular if it is an interior vertex whose incident edges lie on two lines, a boundary vertex with one incident triangle, or a boundary vertex whose incident edges lie on two lines.
- (M1') Every vertex of Γ_h has at least three incident triangles. The angle of Ω_h at a vertex of Γ_h is $\pi + O(h_\Gamma)$, so with two triangles $\Theta(z) = O(h_\Gamma) \rightarrow 0$, contradicting (M2).
- (M2) $\Theta(z) := \max_j |\sin(\theta_j + \theta_{j+1})| \geq \Theta_0 > 0$ at every vertex z , in the notation of Guzmán–Scott [5]. At boundary vertices the wrap-around term is omitted.
- (D) $\tilde{f}$ is smooth and bounded on a fixed neighbourhood of $\overline{\Omega}$, $\tilde{f} = f$ on Ω , and $h \leq h_*$ is small (hence so is $h_\Gamma \leq h$).

Spaces. Fix $k \geq 4$. V_h is the space of continuous piecewise- P_k vector fields, $V_h^R := \{v \in V_h : v|_{\partial R} = 0\}$ and $V_h^0 := \{v \in V_h : v|_{\partial \Omega_h} = 0\}$. Π_h is the space of discontinuous piecewise- P_{k-1} functions and $Z_h := \{v \in V_h^R : \text{div } v = 0\}$. $W_h := \{z \in V_h : z|_{\partial R} = g_I, \text{div } z = 0\}$, where g_I is the P_k Lagrange interpolant of g . Gjerde and Scott, and all computations in Section 9, use $k = 4$.

Forms. n is the outward unit normal of Ω_h on Γ_h (pointing into P_h), and $\langle \cdot, \cdot \rangle$ is the $L^2(\Gamma_h)$ pairing. Let

$$\begin{aligned} a_h(u, v) &= \frac{1}{2}(Du, Dv), \quad Du = \nabla u + (\nabla u)^\top, \\ N_h(u, v) &= a_h(u, v) - \langle \partial_n u, v \rangle - \langle u, \partial_n v \rangle + \frac{\mu}{h} \langle u, v \rangle, \quad \partial_n u := (\nabla u)n. \end{aligned}$$

Definition 2.2 (Method GS [4, (28)], as printed). Find $u_h \in V_h$ with $u_h|_{\partial R} = g_I$ and $p_h \in \Pi_h$ such that

$$N_h(u_h, v) - (p_h, \text{div } v) = (\tilde{f}, v) \quad \forall v \in V_h^R, \quad (q, \text{div } u_h) = 0 \quad \forall q \in \Pi_h.$$

Definition 2.3 (Method CNS*). As GS, with momentum row

$$N_h(u_h, v) - (p_h, \text{div } v) + \langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle = (\tilde{f}, v), \quad \bar{p}_\Gamma(q) := |\Gamma_h|^{-1} \int_{\Gamma_h} q.$$

Exact solution and extensions. (u, p) is smooth with $-\Delta u + \nabla p = f$ and $\operatorname{div} u = 0$ in Ω , $u|_\Gamma = 0$ and $u|_{\partial R} = g$; note $-\operatorname{div} D(u) = -\Delta u$. The stream function ψ with $u = \operatorname{curl} \psi = (\partial_y \psi, -\partial_x \psi)$ is single-valued because $u \cdot n = 0$ on Γ . Let $\tilde{\psi}$ be a smooth extension of ψ to a neighbourhood of $\bar{\Omega}$ and $\tilde{u} := \operatorname{curl} \tilde{\psi}$, so $\operatorname{div} \tilde{u} = 0$. Let $\tilde{p}$ be a smooth extension of p . Then $F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$ vanishes on Ω and is supported in S_h .

Norms and constants. For v with traces on Γ_h ,

$$\|v\|^2 := \|\nabla v\|_{\Omega_h}^2 + \frac{\mu}{h} \|v\|_{\Gamma_h}^2 + \sum_{e \in \mathcal{E}_h^\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2.$$

This is used for discrete fields and for errors $\tilde{u} - z$; since $\tilde{u}$ is smooth, every trace exists. We write $G := \|p - \bar{p}\|_{L^2(\Gamma)}$, with $\bar{p}$ the mean of p over Γ .

C and c depend on (M0)–(M2), k , L and norms of $(u, p, \tilde{\psi})$. They are independent of $h, h_\Gamma, \mu, \gamma, \rho$ unless displayed. C_p is a constant that may also depend on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$. Where G is tracked explicitly, “ δ small” means $\delta \leq \delta_0(C_p)$.

3. GEOMETRY

Lemma 3.1 (Chords). *Let e be a chord of length ℓ and $x \in e$. Then $\operatorname{dist}(x, \Gamma) \leq \ell^2/8 + O(\ell^4) \leq \ell^2/4$. Let s be the signed arclength from the chord midpoint m_e , x^* the nearest point of Γ , and τ_e the unit tangent of e . Then, up to a fixed orientation sign,*

$$n(x^*) - n_e = -s \tau_e + O(\ell^2).$$

The arclength Jacobian of the radial projection $\Gamma_h \rightarrow \Gamma$ is $1 + O(\ell^2)$.

Proof. The sagitta is $1 - \sqrt{1 - \ell^2/4} = \ell^2/8 + \ell^4/128 + \dots$. The normal at x^* is the radial unit vector. Its angle differs from that of n_e by the central angle to x^* , which is $s + O(s^3)$. $\square$

Lemma 3.2 (Uniform trace, Friedrichs and Korn). *There are $C_{\text{tr}}, C_F, \kappa$, independent of h , such that every $v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$ satisfies*

$$\|v\|_{L^2(\Gamma_h)} \leq C_{\text{tr}} \|\nabla v\|, \quad \|v\| \leq C_F \|\nabla v\|, \quad \|\nabla v\|^2 \leq \kappa a_h(v, v).$$

Proof. (1) Write Γ_h in polar form as $r = \rho_h(\theta)$. On a chord with endpoint angles $\theta_m \pm \alpha$ we have $\rho_h = \cos \alpha / \cos(\theta - \theta_m)$. Hence $\|\rho_h - 1\|_\infty \leq Ch_\Gamma^2$ and $\|\rho'_h\|_\infty \leq Ch_\Gamma$.

(2) Let χ be smooth, with $\chi(1) = 1$ and $\chi \equiv 0$ outside a fixed collar in Ω . Define $\Phi_h(r, \theta) = (r + \chi(r)(\rho_h(\theta) - 1), \theta)$.

(3) Φ_h maps Ω onto Ω_h and Γ onto Γ_h , is the identity near ∂R , and is bi-Lipschitz with $\|D\Phi_h - I\|_\infty \leq Ch_\Gamma$ almost everywhere.

(4) For $w = v \circ \Phi_h$ we have $D(w) = D(v) \circ \Phi_h + E$ with $|E| \leq Ch_\Gamma |\nabla v \circ \Phi_h|$.

(5) On the fixed domain Ω , w vanishes on ∂R , a set of positive measure. So the first Korn inequality, the trace inequality and the Friedrichs inequality hold.

(6) Changing variables back gives $\|\nabla v\| \leq C(\|D(v)\| + h_\Gamma \|\nabla v\|)$; absorb the last term for $h_\Gamma \leq h_*$. The other two inequalities transfer the same way. $\square$

Lemma 3.3 (Extension). $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$, so $\|\tilde{u}\|_{L^2(\Gamma_h)} \leq Ch_\Gamma^2$. For $v \in V_h^R$, $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$.

Proof. The first bound follows from Lemma 3.1 and $\tilde{u}|_\Gamma = 0$. For the second, apply a one-dimensional Poincaré inequality across strips of width at most Ch_Γ^2 , and use $|S_h| \leq Ch_\Gamma^2$, $\|F\|_\infty \leq C$ (by (D)) and Lemma 3.2. $\square$

Lemma 3.4 (Wall structure). *On Γ : $\nabla u = \partial_n u \otimes n$, $\partial_n u \cdot n = 0$ and $(\nabla u)^\top n = 0$.*

Proof. The tangential derivatives vanish, and $\operatorname{div} u = n \cdot \partial_n u$. $\square$

Lemma 3.5 (Transposed-gradient term). *For $v \in V_h^R$,*

$$|\langle (\nabla \tilde{u})^\top n, v \rangle| \leq C \sum_{e \in \mathcal{E}_h^\Gamma} (|e|^2 \|v\|_{L^1(e)} + |e|^2 \|\partial_\tau v\|_{L^1(e)}) \leq Ch_\Gamma^{3/2} \|\nabla v\|.$$

The rate $h_\Gamma^{3/2}$ is sharp on V_h^R : a first-layer field with odd trace on each chord attains it, and the dual-norm test of Section 9 gives 1.56–1.57. For the method itself, which is tested on Z_h , sharpness is supported by the CNS* H^1 rates of Table 1, which decrease monotonically toward $3/2$.

Proof. (1) Taylor-expand about x^* and apply Lemmas 3.1 and 3.4:

$$(\nabla \tilde{u}(x))^\top n_e = n(x^*) (\partial_n u(x^*) \cdot (n_e - n(x^*))) + O(h_\Gamma^2) = s a_e + O(h_\Gamma^2),$$

where $a_e := n(m_e^*) (\partial_n u(m_e^*) \cdot \tau_e)$ is constant on e . The $O(h_\Gamma^2)$ remainder collects $|x - x^*|$, the $O(\ell^2)$ term of Lemma 3.1, and $s \cdot O(s)$.

(2) Since $\int_e s ds = 0$,

$$\left| \int_e s a_e \cdot v \right| = \left| \int_e s a_e \cdot (v - v(m_e)) \right| \leq \frac{|e|^2}{4} \|\partial_\tau v\|_{L^1(e)}.$$

(3) Use $\|\cdot\|_{L^1(e)} \leq |e|^{1/2} \|\cdot\|_{L^2(e)}$ and the inverse trace inequality $|e| \|\partial_\tau v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|_{T_e}^2$.

(4) Summing, $\sum_e |e|^{5/2} \|\partial_\tau v\|_{L^2(e)} \leq Ch_\Gamma^2 (\#\mathcal{E}_h^\Gamma)^{1/2} \|\nabla v\| \leq Ch_\Gamma^{3/2} \|\nabla v\|$. The zeroth-order term is handled the same way, finishing with Lemma 3.2. $\square$

4. DISCRETE TOOLS

Lemma 4.1 (Inverse traces). *For $v \in V_h$,*

$$\sum_{e \in \mathcal{E}_h^\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|^2, \quad \text{hence} \quad \|\partial_n v\|_{\Gamma_h} \leq C_I (\rho/h)^{1/2} \|\nabla v\|.$$

For $q \in \Pi_h$,

$$\|q\|_{\Gamma_h}^2 \leq C_I^2 \sum_{T \in \mathcal{T}_h^\Gamma} |e_T|^{-1} \|q\|_T^2 \leq C_I^2 (\rho/h) \|q\|^2.$$

A triangle with two edges on Γ_h only doubles C_I .

Lemma 4.2 (Coercivity and continuity). *Let $\mu_0 := 4\kappa\rho C_I^2$. The condition $\gamma \geq \gamma_0 := 4\kappa C_I^2/c_0$ is sufficient for $\mu \geq \mu_0$. If $\mu \geq \mu_0$, then*

$$N_h(v, v) \geq c_1 \|v\|^2 \quad \forall v \in V_h^R, \quad |N_h(w, v)| \leq M \|w\| \|v\|,$$

with $c_1 = 1/\max(2\kappa(1+C_I^2), 2)$ and $M = 3 + (\kappa C_I^2)^{-1/2}$. Both constants are independent of h , μ and ρ .

Proof. (1) By Lemma 4.1 and Young's inequality with $\varepsilon = 1/(2\kappa)$,

$$2|\langle \partial_n v, v \rangle| \leq 2C_I \|\nabla v\| (\rho/h)^{1/2} \|v\|_{\Gamma_h} \leq \frac{1}{2} a_h(v, v) + 2\kappa\rho C_I^2 h^{-1} \|v\|_{\Gamma_h}^2.$$

(2) Hence $N_h(v, v) \geq \frac{1}{2} a_h(v, v) + (\mu - 2\kappa\rho C_I^2) h^{-1} \|v\|^2$. For $\mu \geq \mu_0$ this is at least $(2\kappa)^{-1} \|\nabla v\|^2 + (\mu/2h) \|v\|^2$.

(3) Adding $\sum_e |e| \|\partial_n v\|^2 \leq C_I^2 \|\nabla v\|^2$ gives the coercivity constant.

(4) For continuity, $a_h(w, v) \leq 2 \|\nabla w\| \|\nabla v\|$. Each ∂_n -term is at most $(\rho/\mu)^{1/2} \|w\| \|v\| \leq (4\kappa C_I^2)^{-1/2} \|w\| \|v\|$. $\square$

Lemma 4.3 (Divergence range). *Under (M1), $\operatorname{div} V_h^R = \Pi_h$ and $W_h \neq \emptyset$. The velocities of GS and CNS* are pointwise divergence-free.*

- Proof.* (1) $\operatorname{div} V_h^0 = \Pi_h \cap L_0^2$ by the inf-sup condition of Lemma 4.5 below (Guzmán–Scott [5], extending Scott–Vogelius [10]). This holds on the doubly connected domain Ω_h ; there is no circularity, since Lemma 4.5 does not use the present lemma.
- (2) The edge bubble $b = \beta_e n_e$ on one $e \subset \Gamma_h$ lies in V_h^R and has $\int \operatorname{div} b = \int_{\Gamma_h} b \cdot n = |e|/6 \neq 0$. This adds the constants, so $\operatorname{div} V_h^R = \Pi_h$.
- (3) For any $\mathcal{G} \in V_h$ with trace g_I , we have $\operatorname{div} \mathcal{G} \in \Pi_h = \operatorname{div} V_h^R$. Hence $\mathcal{G} - v \in W_h$ for a suitable $v \in V_h^R$.
- (4) For the discrete velocity, $u_h - \mathcal{G} \in V_h^R$, and the continuity row forces $\operatorname{div} u_h = 0$. $\square$

Remark 4.4. If a vertex of ∂R is singular (for example a corner with one triangle), then $A_z(\operatorname{div} \mathcal{G}) = 0$ is a condition on g_I that generic data violate, and $W_h = \emptyset$.

Lemma 4.5 (Uniform inf-sup constant). *There is $\beta > 0$, depending only on k , Θ_0 , shape regularity and L , such that for every $q \in \Pi_h \cap L_0^2(\Omega_h)$ there exists $v \in V_h^0$ with $\operatorname{div} v = q$ and $\|\nabla v\| \leq \beta^{-1} \|q\|$.*

Proof. Guzmán and Scott [5, Theorem 1] prove the inf-sup condition for $(V_h^0, \Pi_h \cap L_0^2)$ for $k \geq 4$ under (M1)–(M2). Their proof needs neither quasi-uniformity, nor small h , nor an interior vertex in each triangle. The discrete Fortin operator is built locally from vertex and edge patches, using shape regularity and (M2) only. The domain enters only through the continuous inf-sup constant (via the P_2 – P_0 Bernardi–Raugel pair) and a Poincaré constant. The Poincaré constant is uniform by Lemma 3.2, so it remains to show that the continuous inf-sup constant is uniform over the domains Ω_h .

- Cover Ω_h by the h -dependent pieces $W_j \cap \Omega_h$, where the W_j are fixed angular wedges of half-angle $\alpha < \pi/2$ about the origin. Each $W_j \cap R$ is convex, and $W_j \cap \Omega_h = (W_j \cap R) \setminus P_h$.
- Each $W_j \cap \Omega_h$ is star-shaped with respect to a ball $B(c_j, r_B) \subset W_j \cap R$ with $|c_j| = r_b$, provided every chord line meeting W_j leaves the ball on its outer side. This holds uniformly when $(r_b - r_B) \cos(\alpha + \arcsin(r_B/r_b) + h_\Gamma) > 1$, which can be arranged with fixed α, r_b, r_B because $P_h \subset D$.
- The pieces have diameters, ball radii and pairwise overlap measures bounded above and below independently of h .

Galdi’s Lemma III.3.4 [3] then gives a Bogovskii constant, and hence a continuous inf-sup constant, independent of h . The equivalence of the inf-sup condition with a bounded right inverse of div gives the statement. $\square$

Lemma 4.6 (Approximation in W_h). *Let $k \geq 4$. Let $\sigma_k := k + 2$ for even k and $\sigma_k := k + 1$ for odd k . Then*

$$E_h := \inf_{z \in W_h} \|\tilde{u} - z\| \leq C\varepsilon_h, \quad \varepsilon_h := (1 + (\gamma h_\Gamma)^{1/2}) h^k + h^{\sigma_k} (h_\Gamma^{-1/2} + (\mu/h)^{1/2}).$$

The second term comes from the net flux of the interpolated boundary data. It cannot be removed for the method as posed. Let ν be the outward normal of R and $m_R := \int_{\partial R} g_I \cdot \nu$. Every $z \in W_h$ (including the discrete velocity) has $\int_{\Gamma_h} z \cdot n = -m_R$, while $\int_{\Gamma_h} \tilde{u} \cdot n = 0$. Hence $\|\tilde{u} - z\| \geq (\mu/h)^{1/2} |m_R|/|\Gamma_h|^{1/2}$, and $|m_R| \asymp h^{\sigma_k}$ for generic g .

For bounded γ the term is at most Ch^k whenever $h_\Gamma \geq h^{2(\sigma_k - k)}$: that is, $h_\Gamma \geq h^4$ for even k and $h_\Gamma \geq h^2$ for odd k . It vanishes if g_I is flux-corrected so that $m_R = 0$, and for the test data of Section 9, where $g \cdot \nu$ is a polynomial of degree at most 5 on each side and the flux integrals are exact.

Proof. (1) *Interpolation.* Let $w_h := I_h \tilde{u} \in V_h$ be the Lagrange interpolant of $\tilde{u}$, which is smooth on Ω_h . On ∂R , $w_h = g_I$. Boundary triangles have diameter at most Ch_Γ by shape regularity, so

$$\|\nabla(\tilde{u} - w_h)\| \leq Ch^k, \quad \|\tilde{u} - w_h\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^{k+1}, \quad \|\partial_n(\tilde{u} - w_h)\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k.$$

Hence $\|\tilde{u} - w_h\| \leq C(h^k + (\mu/h)^{1/2}h_\Gamma^{k+1} + h_\Gamma^k) \leq C(1 + (\gamma h_\Gamma)^{1/2})h^k$. Moreover $\operatorname{div} w_h = \operatorname{div}(w_h - \tilde{u}) \in \Pi_h$, with $\|\operatorname{div} w_h\| \leq Ch^k$.

- (2) *Net flux.* Let ν be the outward normal of R . Since $\tilde{u}|_{\partial R} = g$ and $\operatorname{div} \tilde{u} = 0$, we have $\int_{\Gamma_h} \tilde{u} \cdot n = -\int_{\partial R} g \cdot \nu = 0$. Therefore

$$m := \int_{\Omega_h} \operatorname{div} w_h = \int_{\partial R} (g_I - g) \cdot \nu + \int_{\Gamma_h} (w_h - \tilde{u}) \cdot n, \quad |m| \leq Ch^{\sigma_k}.$$

Here equispaced P_k Lagrange interpolation on an edge integrates like the closed $(k+1)$ -point Newton–Cotes rule. That rule is exact for polynomials of degree $\sigma_k - 1$, so each edge contributes $O(|e|^{\sigma_k+1})$.

- (3) *Removing the mean.* Let $b := \sum_{e \in \mathcal{E}_h^\Gamma} \beta_e n_e$, where β_e is the quadratic edge bubble of the boundary triangle on e . Then $b \in V_h^R$, $\int \operatorname{div} b = \int_{\Gamma_h} b \cdot n = |\Gamma_h|/6$, and with $a := 6m/|\Gamma_h|$,

$$\|ab\| \leq C|a|(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) \leq Ch^{\sigma_k}(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) \leq C\varepsilon_h,$$

using $\#\mathcal{E}_h^\Gamma \leq C/h_\Gamma$. Also $\|\operatorname{div}(ab)\| \leq C|a|h_\Gamma^{-1/2} \leq C\varepsilon_h$.

- (4) *Divergence correction.* The field $q := \operatorname{div}(w_h - ab) \in \Pi_h$ has mean zero and $\|q\| \leq C\varepsilon_h$. Lemma 4.5 gives $v \in V_h^0$ with $\operatorname{div} v = q$ and $\|\nabla v\| \leq \beta^{-1}\|q\| \leq C\varepsilon_h$. Since $v = 0$ on Γ_h , $\|v\|^2 = \|\nabla v\|^2 + \sum_e |e| \|\partial_n v\|_{L^2(e)}^2 \leq (1 + C_I^2) \|\nabla v\|^2$.

- (5) Set $z := w_h - ab - v$. Then $z|_{\partial R} = g_I$ and $\operatorname{div} z = 0$, so $z \in W_h$, and

$$\|\tilde{u} - z\| \leq \|\tilde{u} - w_h\| + \|ab\| + \|v\| \leq C\varepsilon_h. \quad \square$$

Remark 4.7. Version 1 of this preprint proved Lemma 4.6 for $k = 4$ by matching Argyris stream-function data on ∂R . The proof above avoids stream functions and holds for every $k \geq 4$.

5. THE ERROR EQUATION

Lemma 5.1. For $v \in V_h^R$,

$$N_h(\tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle = (\tilde{f}, v) + \mathcal{G}(v),$$

where

$$\mathcal{G}(v) := \langle (\nabla \tilde{u})^\top n, v \rangle - \langle \tilde{u}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{u}, v \rangle - (F, v).$$

Proof. The argument has four steps.

- (1) By symmetry of $D(\tilde{u})$ and since $v = 0$ on ∂R ,

$$a_h(\tilde{u}, v) = (D(\tilde{u}), \nabla v) = -(\operatorname{div} D(\tilde{u}), v) + \langle D(\tilde{u})n, v \rangle.$$

- (2) $\operatorname{div} D(\tilde{u}) = \Delta \tilde{u}$.

- (3) $-\Delta \tilde{u} = \tilde{f} - F - \nabla \tilde{p}$, and $-(\nabla \tilde{p}, v) = (\tilde{p}, \operatorname{div} v) - \langle \tilde{p}, v \cdot n \rangle$.

- (4) $D(\tilde{u})n - \partial_n \tilde{u} = (\nabla \tilde{u})^\top n$. □

Lemma 5.2. For $v \in V_h^R$, $|\mathcal{G}(v)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}\|v\|$. For $v \in V_h^0$, $|\mathcal{G}(v)| \leq Ch_\Gamma^{3/2}\|\nabla v\|$.

Proof. We bound the four terms of $\mathcal{G}(v)$ in turn.

- The transposed term is at most $Ch_\Gamma^{3/2}\|\nabla v\|$ by Lemma 3.5.
- $|\langle \tilde{u}, \partial_n v \rangle| \leq Ch_\Gamma^2 \cdot C_I(\rho/h)^{1/2}\|\nabla v\| \leq Ch_\Gamma^{3/2}\|\nabla v\|$.
- The penalty term is at most $(\mu/h)^{1/2}Ch_\Gamma^2\|v\| = C\gamma^{1/2}h_\Gamma^{3/2}\|v\|$.
- The F term is bounded by Lemma 3.3.

For $v \in V_h^0$, the transposed and penalty terms vanish. □

Corollary 5.3 (GS on Z_h). *Let $v \in Z_h$. Then $(\cdot, \operatorname{div} v) = 0$ and $\int_{\Gamma_h} v \cdot n = 0$, so for every constant c ,*

$$N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v), \quad \mathcal{R}(v) := -\langle \tilde{p} - c, v \cdot n \rangle.$$

6. THE PRINTED METHOD

Throughout this section $\gamma \geq \gamma_0$.

Test field. Let $\varphi_0(\theta) := \int_0^\theta (p - \bar{p})(\theta') d\theta'$, which is periodic, and $\varphi := \chi_1(r)\varphi_0(\theta)$, where $\chi_1 \equiv 1$ near $r = 1$ (so $\chi_1'(1) = 0$) and $\chi_1 \equiv 0$ near ∂R . Then $w := \operatorname{curl} \varphi$ has components $w_r = \partial_\theta \varphi / r$ and $w_\theta = -\partial_r \varphi$, and on Γ

$$w = (p - \bar{p})e_r = -(p - \bar{p})n.$$

Let Π_{k+1} be the C^1 piecewise- P_{k+1} interpolant of Morgan and Scott [7] (for $k = 4$, the Argyris interpolant [1]), which exists because $k+1 \geq 5$; its constants depend on the angles, hence on (M0) and (M2). Since $\varphi \equiv 0$ near ∂R and $h \leq h_*$, $v_\varphi := \operatorname{curl} \Pi_{k+1} \varphi \in Z_h$, and $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$.

Theorem 6.1 (Energy norm: exact rate $1/2$). *The error satisfies the upper bound*

$$\|\tilde{u} - u_h\| \leq C \left[(h_\Gamma/\gamma)^{1/2} G + (1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

If $G > 0$ and $h \leq h_0(G)$, with h_0 independent of $\gamma \geq \gamma_0$, it also satisfies the lower bound

$$\|\tilde{u} - u_h\| \geq (2M)^{-1} (h_\Gamma/\gamma)^{1/2} G - C \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

In Scott's variables, $(h_\Gamma/\gamma)^{1/2} = (h/\mu)^{1/2}$.

Proof of the upper bound. For $z \in W_h$, $e_h := z - u_h \in Z_h$. By Corollary 5.3 and Lemma 4.2,

$$c_1 \|e_h\|^2 \leq N_h(z - \tilde{u}, e_h) + \mathcal{R}(e_h) + \mathcal{G}(e_h).$$

Using the Jacobian bound of Lemma 3.1, $|\mathcal{R}(e_h)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2} \|e_h\|$. Then apply Lemmas 5.2 and 4.6. $\square$

Proof of the lower bound. (1) We have $v_\varphi \cdot n_e = -(p - \bar{p})(x^*) n(x^*) \cdot n_e + O(h^k)$, $\mathcal{R} = -\langle \tilde{p} - \bar{p}, v \cdot n \rangle$ and $n(x^*) \cdot n_e = 1 + O(s^2)$. Hence

$$\mathcal{R}(v_\varphi) = \langle (\tilde{p} - \bar{p})(p - \bar{p})(x^*), n(x^*) \cdot n_e \rangle + O(h^k) = G^2 + O(h_\Gamma^2 + h^k).$$

$$(2) \|v_\varphi\|^2 \leq (\mu/h)(G + Ch_\Gamma^2 + Ch^k)^2 + C.$$

$$(3) \text{ By continuity, } M\|\tilde{u} - u_h\| \geq (|\mathcal{R}(v_\varphi)| - |\mathcal{G}(v_\varphi)|)/\|v_\varphi\|, \text{ and Lemma 5.2 bounds } \mathcal{G}.$$

$$(4) |\mathcal{R}|/\|v_\varphi\| \geq (h/\mu)^{1/2} G(1 - C(h_\Gamma^2 + h^k)/G^2 - C(h_\Gamma^2 + h^k)/G - C(h_\Gamma/\gamma_0)^{1/2}/G), \text{ which is at least } \frac{1}{2}(h/\mu)^{1/2} G \text{ for } h \leq h_0(G). \quad \square$$

The lower bound is carried by the boundary parts of $\|\cdot\|$. By Theorem 6.4, $\|\nabla(\tilde{u} - u_h)\|$ is only $O(h/\mu)$.

Corollary 6.2. *If $G > 0$, $\gamma h_\Gamma \ll G$ and $\varepsilon_h \ll (h_\Gamma/\gamma)^{1/2} G$, then $\|\tilde{u} - u_h\| \asymp (h/\mu)^{1/2}$. The mesh-dependent energy rate is exactly $1/2$.*

Lemma 6.3 (Normal-load cancellation). *Let S be edgewise $W^{1,\infty}$ on Γ_h , continuous at the vertices of Γ_h , with $|S \cdot \tau_e| \leq Ch_\Gamma$ on every $e \in \mathcal{E}_h^\Gamma$. (For instance, S smooth with $S \cdot \tau = 0$ on Γ .) Then for $v \in Z_h$,*

$$|\langle S, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2} \|\nabla v\|) \leq C\|\nabla v\|.$$

Proof. (1) *Tangential part.* By Lemma 4.1 and (M0), $|\langle (S \cdot \tau)\tau, \partial_n v \rangle| \leq Ch_\Gamma \|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{1/2} \|\nabla v\|$.

- (2) *Normal part.* On e , v is a polynomial on T_e with $\operatorname{div} v = 0$, so $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$. With $\sigma_e := S \cdot n_e$,

$$\int_e \sigma_e \partial_n v \cdot n_e = \int_e \partial_\tau \sigma_e (v \cdot \tau_e) - [\sigma_e v \cdot \tau_e]_{\partial e}.$$

- (3) At a polygon vertex x shared by e and e' , the endpoint terms combine to $v(x) \cdot (\sigma_{e'} \tau_{e'} - \sigma_e \tau_e)$, of size at most $C(|e| + |e'|)|v(x)|$.
 (4) $\sum_e |e| \|v\|_{L^\infty(e)} \leq C \sum_e |e|^{1/2} \|v\|_{L^2(e)} \leq C |\Gamma_h|^{1/2} \|v\|_{\Gamma_h}$.
 (5) Finish with Lemma 3.2. $\square$

Numerically (Section 9), the dual norm of $v \mapsto \langle S, \partial_n v \rangle$ stays bounded on Z_h for normal S . It grows like $h_\Gamma^{-1/2}$ on V_h^R (no divergence constraint), and also on Z_h for tangential S . Incompressibility is exactly what the proof uses.

Theorem 6.4 (H^1 seminorm: rate 1). *With $C_p := C(1 + \|p - \bar{p}\|_{W^{1,\infty}(\Gamma)})$,*

$$\|\nabla(\tilde{u} - u_h)\| \leq C_p (h_\Gamma/\gamma) + C \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

Proof. (1) *Split.* For $z \in W_h$, write $z - u_h = \delta_R + \delta_G$, where $\delta_R, \delta_G \in Z_h$ solve

$$N_h(\delta_R, v) = \mathcal{R}(v), \quad N_h(\delta_G, v) = \mathcal{G}(v) - N_h(\tilde{u} - z, v) \quad \forall v \in Z_h.$$

Lax–Milgram on Z_h (Lemma 4.2) gives existence and uniqueness.

- (2) *Geometric part.* $\|\nabla \delta_G\| \leq \|\delta_G\| \leq c_1^{-1} (C(1 + \gamma^{1/2}) h_\Gamma^{3/2} + M E_h)$.
 (3) *Ansatz.* Heuristically $\delta_R \approx -(h/\mu)(p - \bar{p})n = (h/\mu)w$ on Γ_h . Set $S_h := v_\varphi \in Z_h$ and $r_h := \delta_R - (h/\mu)S_h \in Z_h$. Then $N_h(r_h, v) = m(v) + \ell(v)$ with

$$\begin{aligned} m(v) &:= \mathcal{R}(v) - \langle S_h, v \rangle = -\langle (\tilde{p} - \bar{p})n + S_h, v \rangle, \\ \ell(v) &:= -(h/\mu) [a_h(S_h, v) - \langle \partial_n S_h, v \rangle - \langle S_h, \partial_n v \rangle]. \end{aligned}$$

- (4) *Bound on m .* On e ,

$$(\tilde{p} - \bar{p})n_e + S_h = (p - \bar{p})(x^*)(n_e - n(x^*)) + O(h_\Gamma^2) + O(h^k) = (p - \bar{p})(m_e^*) s \tau_e + O(h_\Gamma^2) + O(h^k).$$

The leading term is odd in s with a constant coefficient. The argument of Lemma 3.5 handles it, and the $O(h^k)$ remainder contributes $Ch^k \|v\|_{L^1(\Gamma_h)}$. Hence $|m(v)| \leq C_p (h_\Gamma^{3/2} + h^k) \|\nabla v\|$.

- (5) *Bound on ℓ .* $|a_h(S_h, v)| + |\langle \partial_n S_h, v \rangle| \leq C_p \|\nabla v\|$. By Lemma 6.3 applied to $S = w$,

$$|\langle S_h, \partial_n v \rangle| \leq |\langle w, \partial_n v \rangle| + |\langle S_h - w, \partial_n v \rangle| \leq C_p \|\nabla v\| + Ch^k h_\Gamma^{-1/2} \|\nabla v\|.$$

So $|\ell(v)| \leq (h/\mu)(C_p + Ch^k h_\Gamma^{-1/2}) \|\nabla v\|$.

- (6) *Close.* Testing with $v = r_h$,

$$c_1 \|r_h\|^2 \leq [C_p(h/\mu + h_\Gamma^{3/2} + h^k) + C(h/\mu)h^k h_\Gamma^{-1/2}] \|\nabla r_h\|.$$

The last two terms are at most $C\varepsilon_h$, since $(h/\mu)h^k h_\Gamma^{-1/2} = \gamma^{-1}h^k h_\Gamma^{1/2} \leq Ch^k$. Hence

$$\|\nabla \delta_R\| \leq (h/\mu) \|\nabla S_h\| + \|\nabla r_h\| \leq C_p (h_\Gamma/\gamma) + C(h_\Gamma^{3/2} + \varepsilon_h). \quad \square$$

Remark 6.5 (Why rate 1 and not 1/2). Without incompressibility, Lemma 6.3 would give only $Ch_\Gamma^{-1/2}$. The H^1 error would then be $(h/\mu)h_\Gamma^{-1/2}$: rate 1/2, with element-scale oscillation in the trace. The inconsistency $-pn$ is purely normal, and $\operatorname{div} v = 0$ converts $\partial_n v \cdot n$ into a tangential derivative. That is why GS still converges at rate 1 in H^1 .

Theorem 6.6 (Boundary slip; H^1 lower bound). *Let $\gamma \geq \gamma_0$ and $G > 0$, and put $X := C_p(h_\Gamma/\gamma) + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h$, the right-hand side of Theorem 6.4. Suppose that, for a fixed small δ ,*

$$X \leq \delta G \min(G, h_\Gamma^{1/2}), \quad h_\Gamma/\gamma \leq \delta G, \quad h_\Gamma \leq h_0(G).$$

All three hold for fixed γ and bounded ρ as $h_\Gamma \rightarrow 0$. Then

- (i) $c(h_\Gamma/\gamma)G \leq \|\tilde{u} - u_h\|_{L^2(\Gamma_h)} \leq C(h_\Gamma/\gamma)^{1/2} \|\tilde{u} - u_h\|$;
- (ii) $\|\nabla(\tilde{u} - u_h)\| \geq c(h_\Gamma/\gamma)G - Ch^{k+1/2}$.

Combined with Theorem 6.4, $\|\nabla(\tilde{u} - u_h)\| \asymp h/\mu$ in this regime.

Proof of (i). Let $e := \tilde{u} - u_h$ and $v := v_\varphi$. Corollary 5.3 rearranges to

$$\frac{\mu}{h} \langle e, v \rangle = \mathcal{R}(v) + \mathcal{G}(v) - a_h(e, v) + \langle \partial_n e, v \rangle + \langle e, \partial_n v \rangle.$$

We bound the terms, using that v is smooth up to $O(h^k)$.

- (1) $\mathcal{R}(v) = G^2 + O(h_\Gamma^2 + h^k)$.
- (2) $|\mathcal{G}(v)| \leq C(h_\Gamma^{3/2} + \gamma h_\Gamma^3 + \gamma h_\Gamma h^k)$. On the chord, $\tilde{u} = \pm d \partial_n u(x^*) + O(d^2)$ is tangential while $v \approx -(p - \bar{p})n$ is normal. Hence $\tilde{u} \cdot v = O(h_\Gamma^4 + h_\Gamma^2 h^k)$, the second term coming from $v_\varphi - w$.
- (3) $|a_h(e, v)| \leq C_p \|\nabla e\| \leq C_p X \leq C_p \delta G^2$, by Theorem 6.4.
- (4) By Theorem 6.1 and $h/\mu = h_\Gamma/\gamma \leq \delta G$,

$$|\langle e, \partial_n v \rangle| \leq C_p \|e\|_{\Gamma_h} \leq C_p (h/\mu)^{1/2} \|e\| \leq C_p [(h/\mu)G + X] \leq 2C_p \delta G^2.$$

(5) We have

$$|\langle \partial_n e, v \rangle| \leq \|\partial_n(\tilde{u} - z)\|_{\Gamma_h} \|v\| + C_I(\rho/h)^{1/2} (\|\nabla(z - \tilde{u})\| + \|\nabla e\|) \|v\|_{\Gamma_h}.$$

By Theorem 6.4 and $(\rho/h)^{1/2} \leq (c_0 h_\Gamma)^{-1/2}$, this is at most $Ch_\Gamma^{-1/2} XG \leq C\delta G^2$.

For δ small, $|\langle e, v \rangle| \geq \frac{1}{2}(h/\mu)G^2$. Dividing by $\|v\|_{\Gamma_h} \leq G + Ch_\Gamma^2$ gives the lower bound. The upper bound is $\|e\|_{\Gamma_h} \leq (h/\mu)^{1/2} \|e\|$. $\square$

Proof of (ii). Let $E \in H^1(\Omega_h)$ lift $(g - g_I)|_{\partial R}$, supported near ∂R , with $E|_{\Gamma_h} = 0$ and $\|E\|_{H^1} \leq C\|g - g_I\|_{H^{1/2}(\partial R)} \leq Ch^{k+1/2}$. By Lemma 3.2, $\|e\|_{\Gamma_h} = \|e - E\|_{\Gamma_h} \leq C_{\text{tr}}(\|\nabla e\| + \|\nabla E\|)$. $\square$

Corollary 6.7 (Exact leading term). *Let $G > 0$ and $h_\Gamma \rightarrow 0$ with $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$ and $\varepsilon_h/(h_\Gamma/\gamma)^{1/2} \rightarrow 0$. Then*

$$\tilde{u} - u_h = (h/\mu) v_\varphi + \vartheta_h, \quad \|\vartheta_h\| = o((h/\mu)^{1/2}), \quad \|\vartheta_h\|_{\Gamma_h} = o(h/\mu),$$

and consequently

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1.$$

On Γ_h , $v_\varphi \approx -(p - \bar{p})n$. So to leading order the printed method replaces the no-slip condition by the leak condition $(\mu/h) u_h \cdot n \approx p - \bar{p}$: the penalty, not a traction, balances the pressure.

Proof. Write $\tilde{u} - u_h = (\tilde{u} - z) + \delta_R + \delta_G$ in the notation of Theorem 6.4, with $\delta_R = (h/\mu)S_h + r_h$ and $S_h = v_\varphi$.

(1) From step 6 of Theorem 6.4,

$$c_1 \|r_h\|^2 \leq [C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h] \|r_h\|.$$

So $\|r_h\| \leq C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h$ and $\|r_h\|_{\Gamma_h} \leq (h/\mu)^{1/2} \|r_h\|$.

(2) $\|\delta_G\| + \|\tilde{u} - z\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$, and the Γ_h norm carries an extra factor $(h/\mu)^{1/2}$.

(3) Relative to $(h/\mu)^{1/2}G$ (energy) and $(h/\mu)G$ (slip), steps 1–2 contribute

$$O((h/\mu)^{1/2}/G + (\gamma^{1/2} + \gamma)h_\Gamma/G + \varepsilon_h(\gamma/h_\Gamma)^{1/2}/G) \rightarrow 0.$$

(4) By Lemma 3.1 (Jacobian and displacement of Γ_h from Γ), $\|v_\varphi\|_{\Gamma_h} = G(1 + O(h_\Gamma^2)) + O(h^k)$ and $\|v_\varphi\|^2 = (\mu/h)\|v_\varphi\|_{\Gamma_h}^2 + O(1)$. $\square$

This gives the sharp answer to Scott's question for the printed method. The estimate $h_\Gamma^{3/2} + h^k$ fails exactly when p is not constant on Γ . The error is then the explicit field $(h/\mu)v_\varphi$, with the universal constant 1 in both the slip norm and the energy norm.

The H^1 constant $\|\nabla(\tilde{u} - u_h)\|_{\mu/(hG)}$ is not universal. It depends on the global Stokes response to the leak, since r_h has the same order as $(h/\mu)S_h$ in H^1 . In the computations it is about 2.7–2.8 (Table 3).

7. THE PRESSURE-CONSISTENT METHOD

Proposition 7.1 (The naive consistent form is singular). *Replace $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$ by $\langle p_h, v \cdot n \rangle$. The resulting linear system is square ($V_h^R \times \Pi_h$ tested by $V_h^R \times \Pi_h$). It has the kernel vector $(0, 1)$ and is therefore singular.*

Proof. For $v \in V_h^R$,

$$-(1, \operatorname{div} v) + \langle 1, v \cdot n \rangle = - \int_{\partial\Omega_h} v \cdot n + \int_{\Gamma_h} v \cdot n = 0.$$

The continuity row vanishes at $u = 0$. $\square$

The system is solvable only for right-hand sides orthogonal to a left null vector, which this argument does not identify. With generic data (Section 9), an iterated-penalty solver and a bordered direct solve both returned discrete velocities that are not divergence-free (discrete divergence about $4.5 \cdot 10^{-3}$). This is consistent with incompatibility.

Let β be the uniform inf-sup constant of Lemma 4.5.

Theorem 7.2 (CNS*: well posed, with Scott's rate). *There is $\gamma_2 \asymp \max(\gamma_0, \beta^{-2})$ such that for $\gamma \geq \gamma_2$ the method CNS* has a unique solution and*

$$\|\tilde{u} - u_h\| \leq C \left[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h \right].$$

Consequently, for every fixed $k \geq 4$, for γ in a bounded range $[\gamma_2, \gamma_{\max}]$ and $h_\Gamma \geq h_\Omega^{2(\sigma_k - k)}$ (that is, $h_\Gamma \geq h_\Omega^4$ for even k and $h_\Gamma \geq h_\Omega^2$ for odd k),

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k).$$

This condition ensures that the flux term of ε_h is at most Ch^k . Without it, that single extra term is kept, and by the remark after Lemma 4.6 it is then genuinely present unless g_I is flux-corrected. This is the rate Scott conjectured, for general Stokes data. The L^2 part of the H^1 norm uses the lift E from the proof of Theorem 6.6(ii).

Proof. (1) *Consistency.* Write $B^*(q, v) := -(q, \operatorname{div} v) + \langle q - \bar{p}_\Gamma(q), v \cdot n \rangle$.

- For a constant c , $B^*(q + c, v) = B^*(q, v) - c \int_{\Gamma_h} v \cdot n$.
- $B^*(\tilde{p}, v)$ equals the pressure form of Lemma 5.1 minus $\bar{p}_\Gamma(\tilde{p}) \int_{\Gamma_h} v \cdot n$.
- Hence $p^* := \tilde{p} - \bar{p}_\Gamma(\tilde{p})$ satisfies $N_h(\tilde{u}, v) + B^*(p^*, v) = (\tilde{f}, v) + \mathcal{G}(v)$ for all $v \in V_h^R$.

With $e_u := \tilde{u} - u_h$ and $e_p := p^* - p_h$ this gives

$$N_h(e_u, v) + B^*(e_p, v) = \mathcal{G}(v) \quad \forall v \in V_h^R. \quad (1)$$

- (2) *Split.* Write $e_p = \eta + \xi$ with $\eta := p^* - \pi_h p^*$ and $\xi \in \Pi_h$. For $v \in V_h^0$, $(\eta, \operatorname{div} v) = 0$ because $\operatorname{div} V_h^0 \subset \Pi_h$.
- (3) *Pressure.* For $v \in V_h^0$, (1) gives

$$(\xi - \bar{\xi}, \operatorname{div} v) = N_h(e_u, v) - \mathcal{G}(v) = a_h(e_u, v) - \langle e_u, \partial_n v \rangle - \mathcal{G}(v).$$

Using $(\rho/\mu)^{1/2} \leq C$ and Lemma 5.2, the right-hand side is at most $C(\|e_u\| + h_\Gamma^{3/2})\|\nabla v\|$. Lemma 4.5 then gives $\beta\|\xi - \bar{\xi}\| \leq C(\|e_u\| + h_\Gamma^{3/2})$.

- (4) *Velocity.* For $z \in W_h$, $e_h := z - u_h \in Z_h$. Test (1) with $v = e_h$. Since $(e_p, \operatorname{div} e_h) = 0$ and $\langle \bar{p}_\Gamma(e_p), e_h \cdot n \rangle = 0$,

$$c_1 \|e_h\|^2 \leq M E_h \|e_h\| + |\mathcal{G}(e_h)| + |\langle \eta, e_h \cdot n \rangle| + |\langle \xi - \bar{\xi}, e_h \cdot n \rangle|,$$

where

$$|\langle \eta, e_h \cdot n \rangle| \leq C h_\Gamma^k (h/\mu)^{1/2} \|e_h\|,$$

$$|\langle \xi - \bar{\xi}, e_h \cdot n \rangle| \leq C_I (\rho/h)^{1/2} \|\xi - \bar{\xi}\| (h/\mu)^{1/2} \|e_h\| \leq C(c_0 \gamma)^{-1/2} \|\xi - \bar{\xi}\| \|e_h\|.$$

- (5) *Absorb.* Insert step 3 and $\|e_u\| \leq E_h + \|e_h\|$. If $\gamma \geq \gamma_2$ with $C(c_0 \gamma_2)^{-1/2} \beta^{-1} \leq c_1/2$, the pressure term is absorbed.
- (6) *Uniqueness.* Apply steps 3–5 to the homogeneous problem ($g_I = 0$, so $z = 0 \in W_h$). This gives $u_h = 0$, and then $p_h = c$ is constant. Now (1) reads $-c \int_{\Gamma_h} v \cdot n = 0$ for all v , and the edge bubble of Lemma 4.3 gives $c = 0$. The system is square, so uniqueness gives existence. $\square$

8. THE PENALTY THRESHOLD SCALES WITH ρ

Proposition 8.1.

- (a) Sufficiency. $\mu \geq \mu_0 = 4\kappa C_I^2 \rho$ implies coercivity (Lemma 4.2).
- (b) Necessity, with an exact certificate. Let z be a vertex of a straight part of the Nitsche boundary, and let ℓ_z be the length of its boundary edges. Suppose its star, dilated by the factor ℓ_z , is congruent to one of the reference stars $\mathcal{S}_3$ or $\mathcal{S}'_4$ below. If $\mu/(h/\ell_z) < 9$, then N_h is indefinite on Z_h .

The same holds, by continuity, at a vertex of Γ_h with interior angle $\pi + O(h_\Gamma)$ whose star is a small perturbation of $\mathcal{S}_3$ or $\mathcal{S}'_4$. This uses that both reference stars are non-singular: every vertex, including the rim vertices, has non-collinear incident edges. The dimension of the divergence-free subspace therefore equals the generic count and is locally constant. This fails at the singular $m = 2$ configuration, which (M1') excludes.

So, for meshes that contain such a boundary star at a vertex with $\rho \asymp h/\ell_z$, the condition $\mu_0 \asymp \rho$ is both necessary and sufficient. The threshold is local: what is required is $\mu/h \gtrsim c/\ell_z$. The ρ -scaling is the price of Scott's normalisation of the penalty by the global h .

Proof of (b). (1) In two dimensions, $a(v, v)$ and $\langle \partial_n v, v \rangle$ are invariant under dilation, and $\|v\|_{L^2(\Gamma_h)}^2$ scales linearly.

- (2) A witness v on the reference star, extended by zero, lies in Z_h : it is continuous, vanishes on the star's outer chords and on ∂R , and is divergence-free.
- (3) It satisfies $N_h(v, v) = a(v, v) - 2\langle \partial_n v, v \rangle + (\mu \ell_z/h) \|v\|_{\text{ref}}^2$.
- (4) By the certificate below, $2B - A - 9C > 0$ on each reference star, so $N_h(v, v) < 0$ when $\mu \ell_z/h < 9$. $\square$

The exact certificate. The reference stars are fans at $O = (0, 0)$ with rational rim points:

- $\mathcal{S}_3$: rim points $(1, 0), (1, 1), (-1, 1), (-1, 0)$;
- $\mathcal{S}'_4$: rim points $(1, 0), (1, 1), (0, \frac{6}{5}), (-1, 1), (-1, 0)$.

The apex of $\mathcal{S}'_4$ is moved off the line $y = 1$. With apex $(0, 1)$ the rim vertex is singular, the kernel dimension jumps from 16 to 17, and the perturbation argument does not apply.

The Nitsche boundary is $\{y = 0\}$, and the rim chords carry zero Dirichlet data. The velocity is represented on each triangle by P_4 polynomials in global coordinates with rational coefficients. The computation is done in exact rational arithmetic (SymPy):

- Continuity across the spokes, vanishing on the rim, and pointwise $\operatorname{div} v = 0$ are imposed as polynomial identities. This gives 80 and 110 linear constraints and kernel dimensions 14 and 16, the generic counts.
- The quadratic forms $A = \sum_T \int \frac{1}{2} Dv : Dv$, $B = \int (\partial_n v) \cdot v$ and $C = \int |v|^2$ are integrated exactly.
- A rational witness in the kernel satisfies $2B - A - 9C > 0$ *exactly* on each star. Its exact Rayleigh quotients $(2B - A)/C$ are $9.97854\dots$ ($\mathcal{S}_3$) and $9.26747\dots$ ($\mathcal{S}'_4$).

Computation: `code/lamstar_exact.py`, `code/lamstar_exact_v2.py`; output `logs/lamstar_exact_*.log`.

The witnesses are continuous piecewise- P_4 and divergence-free, so they lie in Z_h for every $k \geq 4$. The necessity statement therefore holds for all $k \geq 4$; for $k > 4$ the true local threshold is at least as large.

In floating point, the largest Rayleigh quotient λ^* is about 15 on uniform fans with $m = 3, 4$ and ranges over 14.8–38.5 on random 4-fans (`code/lamstar.py`). As a global check, the measured threshold $\gamma^* = \mu^*/\rho$ stays in 18.1–20.1 as ρ varies over 3.3–44 on two mesh levels. Under refinement at fixed ρ it lies in 14.7–20.8, with increments that roughly halve per doubling (Table 4).

9. COMPUTATIONS

Code and meshes. We use P_4 Scott–Vogelius elements with Nitsche’s method on the inscribed polygon and strong Dirichlet data on the square $R = (-2.5, 2.5)^2$. Divergence is enforced exactly: all ten P_3 moments per triangle.

Saddle-point systems are solved by SuperLU [6] with partial pivoting and three steps of iterative refinement. Over the 148 production solves (74 cases, two methods each), the relative residual is at most $3.8 \cdot 10^{-15}$ and the discrete divergence at most $7 \cdot 10^{-11}$ (median $9 \cdot 10^{-12}$).

The code reproduces the zero-pressure-gradient benchmark of [2] to 10^{-5} relative on four meshes. It gives $\mu^* = 60.343$ at $N = 20$, against the published 60.338.

The mesh family has N equal boundary chords and $\rho \approx 3.3$, with h_Ω decreasing from 1.505 to 0.109 for $N = 16 \rightarrow 256$; rates in h_Ω and h_Γ therefore coincide. Rates in the tables are computed between consecutive runs of the full sequence $N = 16, 24, 32, 48, 64, 96, 128, 192, 256$, some of which are omitted from the tables.

Test problems.

- A** Scott’s benchmark: the shear flow $u = (1 - r^{-2})(-y, x)$ with $p \equiv 0$, so $G = 0$.
- B_a** $\psi = (r^2 - 1)^2(x + y^2/2)/10$ and $p = a(x^2y - y + x/2)$, so $G = a(7\pi/8)^{1/2}$: $G = 1.658$ for $a = 1$ and 165.8 for $a = 100$.
- P** Pure pressure: $u \equiv 0$, $f = \nabla p$, with p as in B₁. CNS* reproduces this exactly ($u_h = 0$ and $p_h = p - \bar{p}_\Gamma(p)$, since $p \in \Pi_h$); its computed velocity is below 10^{-13} in H^1 . The GS output is therefore *exactly* the response to the missing traction.
- C** ψ as in B, with the non-polynomial pressure $p = e^{x/2} \cos y$. This exercises the term $p^* - \pi_h p^*$ of Theorem 7.2.

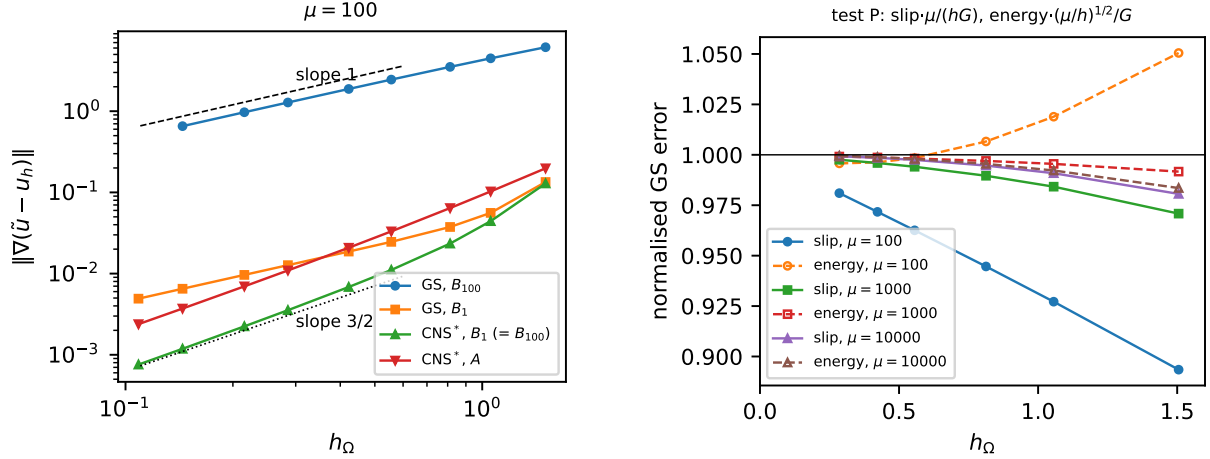


FIGURE 1. Left: H^1 errors at $\mu = 100$. GS has slope 1 as soon as $G > 0$, while CNS* tends to slope 3/2 independently of the pressure. Right: the normalised slip and energy errors of GS on test P converge to 1 (Corollary 6.7).

TABLE 1. CNS* attains Scott's rate $h_\Gamma^{3/2}$ (Theorem 7.2). H^1 error, $\mu = 100$.

N	h_Ω	A	rate	B_1	rate
32	0.813	$6.41 \cdot 10^{-2}$	1.80	$2.34 \cdot 10^{-2}$	2.43
64	0.422	$2.08 \cdot 10^{-2}$	1.69	$6.83 \cdot 10^{-3}$	1.76
96	0.285	$1.09 \cdot 10^{-2}$	1.64	$3.54 \cdot 10^{-3}$	1.68
128	0.215	$6.96 \cdot 10^{-3}$	1.60	$2.24 \cdot 10^{-3}$	1.63
192	0.145	$3.71 \cdot 10^{-3}$	1.58	$1.19 \cdot 10^{-3}$	1.60
256	0.109	$2.39 \cdot 10^{-3}$	1.56	$7.58 \cdot 10^{-4}$	1.57

Tests A and B were run with $\mu \in \{100, 1000, 10000\}$ and $N \leq 192$, plus $N = 256$ for $(A, 100)$ and $(B_1, 100)$, on a 32-core machine (36 minutes in total). Test P (all three μ) and test C ($\mu = 100$) were run with $N \leq 96$ on a 2-core machine.

Theorem 7.2: CNS*. Table 1 gives the CNS* errors. At $N = 256$ the energy-norm rate is 1.52 and the slip rate is 2.02; the slip carries an extra factor $(h/\mu)^{1/2}$. The H^1 rates decrease monotonically toward 3/2, the geometric term of Lemma 3.5. The h_Ω^4 term is invisible at these levels because the geometric term dominates.

The error is independent of the pressure.

- For polynomial pressures this is structural ($p \in \Pi_h$). B_1 and B_{100} agree to all printed digits wherever both were run ($N \leq 192$, every μ).
- With the non-polynomial pressure of test C, the CNS* H^1 error differs from that of B_1 by a relative $7.9 \cdot 10^{-7}$, $3.9 \cdot 10^{-8}$ and $9.7 \cdot 10^{-9}$ at $N = 16, 64$ and 96 . This is the size of the contribution $\langle p^* - \pi_h p^*, v \cdot n \rangle$, which is of higher order, as Theorem 7.2 predicts.

Theorems 6.1–6.4: the printed method. Table 2 compares the H^1 rates of GS and CNS* on B_{100} .

For the non-polynomial test C at $\mu = 100$, the GS H^1 rate from $N = 64$ to 96 is already 1.12, against 1.68 for CNS*. For the small pressure B_1 at $\mu = 1000$, GS still shows 1.51 at $N = 192$.

TABLE 2. GS loses Scott's rate as soon as $G > 0$. H^1 rate between $N = 128$ and $N = 192$, test B₁₀₀.

μ	GS	CNS*
100	0.99	1.60
1000	1.00	1.56
10000	1.06	1.58

TABLE 3. The leading term of Corollary 6.7. Normalised GS error on test P ($G = 1.658$), where $e = \tilde{u} - u_h$.

N	μ	$\ \nabla e\ \mu/(hG)$	$\ e\ _{\Gamma_h} \mu/(hG)$	$\ e\ (\mu/h)^{1/2}/G$
16	100	2.464	0.894	1.051
32	100	2.612	0.945	1.007
64	100	2.684	0.972	0.996
96	100	2.708	0.981	0.996
96	1000	2.770	0.998	0.999
96	10000	2.790	0.9993	0.9994

The $(h/\mu)G$ term dominates only once $(h/\mu)G \gtrsim h_\Gamma^{3/2}$, which always happens eventually because $h_\Gamma^{3/2}$ decays faster. Larger μ or smaller G delays the crossover but does not remove it.

Corollary 6.7: the leading term. Table 3 and Figure 1 (right) show the normalised GS error on test P.

- The slip and energy ratios converge to 1. The defect in the slip ratio shrinks with h/μ : at $N = 96$ it is 0.019, 0.0025 and 0.0007 for $\mu = 10^2$, 10^3 and 10^4 .
- The H^1 constant is not covered by the corollary. For $\mu = 100$ it rises from 2.46 to 2.71; for $\mu = 10^3$ and 10^4 it falls, from 2.78 to 2.77 and from 3.06 to 2.79. This is consistent with a common, μ -independent limit near 2.7–2.8, but the data do not establish one.
- On test B, GS itself (B₁, $\mu = 100$, $N = 256$) gives slip and energy ratios 0.995 and 1.001.
- For $\mu = 10^4$ with the small pressure B₁, $\gamma h_\Gamma \approx 74 \gg G$ at $N = 192$, so the hypothesis of the corollary fails. The GS energy ratio there is about 7.8, dominated by the geometric error rather than the traction response.
- The difference $u_{\text{GS}} - u_{\text{CNS}^*}$ is not a clean proxy for the traction response on test B. It also contains the difference between the two methods on the velocity part, which is $O(h_\Gamma^{3/2})$ and non-zero even for test A. Test P removes this.

Proposition 8.1: the penalty study. The coercivity threshold μ^* on Z_h is the smallest μ for which the unpivoted symmetric factorisation of $A_\mu + rB^\top M^{-1}B$ has only positive pivots; we locate it by bisection. Table 4 and Figure 2 show the results.

Under refinement at fixed ρ , γ^* increases by 3.4 ($N = 8 \rightarrow 16$), 1.8 ($16 \rightarrow 32$) and 0.95 ($32 \rightarrow 64$). The increments roughly halve, which suggests, but does not prove, a limit near 22.

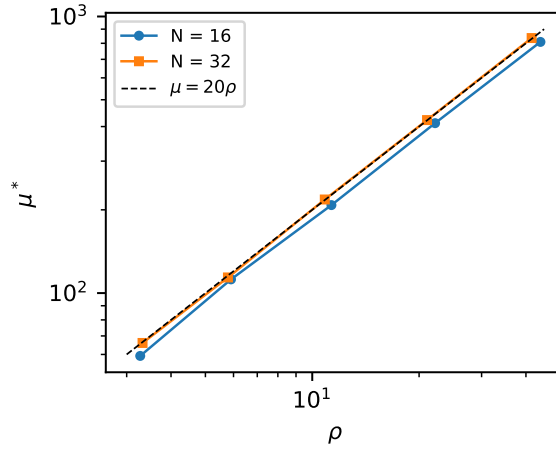
In part (b) of Table 4, ρ is varied by coarsening the far field while the boundary layer is held fixed. Then μ^* is linear in ρ over more than a decade, with a constant slope. The threshold is thus set locally at Γ_h ($\mu/h \gtrsim c/h_\Gamma$), and the ρ -scaling is the cost of normalising the penalty by the global h .

In practical terms:

TABLE 4. Coercivity threshold μ^* and $\gamma^* = \mu^*/\rho$.

<i>(a) Refinement at fixed ρ.</i>								
N	8	12	16	20	24	32	48	64
ρ	3.41	3.02	3.28	3.17	3.30	3.33	3.37	3.40
μ^*	50.3	51.8	59.3	60.3	63.3	66.0	69.1	70.7
γ^*	14.7	17.1	18.1	19.1	19.2	19.9	20.5	20.8

<i>(b) Varying ρ through the outer box $(-L, L)^2$.</i>						
	L	2.5	5	10	20	40
$N = 16$	ρ	3.28	5.90	11.3	22.2	43.9
	μ^*	59.3	112.1	208.0	411.5	809.3
	γ^*	18.1	19.0	18.4	18.5	18.4
$N = 32$	ρ	3.33	5.79	10.8	21.1	41.5
	μ^*	66.0	113.9	218.0	422.2	835.7
	γ^*	19.9	19.7	20.1	20.0	20.1

FIGURE 2. Coercivity threshold μ^* against ρ (Table 4(b)).

- a safe choice is $\mu \geq 25\rho$, that is, $\gamma \geq 25$;
- a fixed μ fails once the far field is coarsened beyond $\rho \approx \mu/20$.

The exact certificate $\lambda \geq 9$ on $\mathcal{S}_3$ and $\mathcal{S}'_4$ is a local lower bound, consistent with the global $\gamma^* \approx 18$ –21.

Lemma tests. We compute exact discrete dual norms, $\sup_v \ell(v)/\|\nabla v\|$ over Z_h or V_h^R , by a saddle-point solve with the pointwise divergence constraint. Table 5 gives the rates for $N = 48 \rightarrow 64$.

Computation: code/lemma_tests.py 16,24,32,48,64; output logs/lemma_tests.log.

APPENDIX A. REPRODUCIBILITY

All code, raw outputs and logs are in the repository that accompanies this paper. Paths below are relative to its root.

TABLE 5. Exact discrete dual norms; rate in h_Γ for $N = 48 \rightarrow 64$ (positive means decay).

Functional	Space	Rate
$\langle S, \partial_n v \rangle, S$ normal	Z_h	+0.05 (bounded, ≈ 1.75)
$\langle S, \partial_n v \rangle, S$ normal	V_h^R	-0.52
$\langle S, \partial_n v \rangle, S$ tangential	Z_h	-0.52
$\langle (\nabla \tilde{u})^\top n, v \rangle$, tests A / B	V_h^R	1.57 / 1.56 (Lemma 3.5: 3/2)

Code (code/).

- `svn.py`: the solver. It contains the mesh generator, the P_4 space, assembly of N_h , B and the CNS* coupling, the SuperLU saddle-point solver (`solve_lu`), error norms, and the exact solutions for tests A, B, P and C.
- `job.py`: one study or threshold job per process.
- `run_all.py`: memory- and CPU-aware scheduler for the full campaign.
- `report.py` and `report_local.py`: produce the tables.
- `plots.py`: produces the figures.
- `penalty.py`: computes the coercivity threshold μ^* .
- `lemma_tests.py`: computes the dual norms of Table 5.
- `lamstar_exact.py` and `lamstar_exact_v2.py`: the exact certificates of Proposition 8.1.
- `lamstar.py`: the floating-point λ^* on other fans.

Where each number comes from.

Result	Command	Output
Tables 1, 2, 4	<code>python3 run_all.py</code>	<code>results/pod/report.txt</code>
Table 3, test C	<code>python3 job.py study N P1 MU and</code> <code>python3 job.py study N C 100,</code> then <code>report_local.py</code>	<code>results/local/report_local.txt</code>
Figures	<code>python3 plots.py</code>	<code>paper/figures/*.pdf</code>
Certificate for S_3	<code>python3 lamstar_exact.py</code>	<code>logs/lamstar_exact_</code> <code>S3_S4singular.log</code>
Certificate for S'_4	<code>python3 lamstar_exact_v2.py</code>	<code>logs/lamstar_exact_S4prime.log</code>
Table 5	<code>python3 lemma_tests.py</code> 16,24,32,48,64	<code>logs/lemma_tests.log</code>
Floating-point λ^*	<code>python3 lamstar.py</code>	<code>logs/lamstar_float.log</code>

Resources. The full A/B campaign (74 study jobs up to $N = 256$ and 16 threshold jobs) took 36 minutes on 32 cores with 125 GB of memory. The largest single job ($N = 256$, both methods) takes about 23 minutes; the scheduler reserved up to 55 GB for it. Every job with $N \leq 96$ runs on a 2-core, 7 GB machine.

The environment was Python 3.11 with NumPy 2.4, SciPy 1.17 (SuperLU) and SymPy 1.14.

USE OF GENERATIVE-AI TOOLS DECLARATION

The author used Anthropic’s Claude Opus 5.5 as a research-support tool. It assisted with the following:

- developing and checking the proofs;
- implementing, running and validating the finite element code and the exact-arithmetic certificates;
- independently reviewing the arguments as a referee;

- preparing the manuscript and the accompanying repository.

The author directed the project and retains authorship of the work. Responsibility for all mathematical statements, computations, references and conclusions rests with the author.

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